TUTORIAL- 7 Answers (EED-201)

Ans-1

(i) $x(t) = \frac{A}{2} + \frac{2A}{\pi} \sum_{m=0}^{\infty} \frac{1}{2m+1} \sin(2m+1)\omega_0 t$

(ii) $\delta_{T_0}(t) = \frac{1}{T_0} + \frac{2}{T_0} \sum_{k=1}^{\infty} \cos k\omega_0 t$; $\omega_0 = 2\pi/T_0$

Ans-2 (i) $x(t) = \frac{A}{2} - \frac{2A}{\pi^2} \sum_{m=-\infty}^{\infty} \frac{1}{(2m+1)^2} e^{j(2m+1)\omega_0 t}$

(ii) $x(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{m=0}^{\infty} \frac{1}{(2m+1)^2} \cos(2m+1)\omega_0 t$

Ans-3 $\omega_2 = \omega_1$;

$b_k = (a_k + a_{-k}) e^{-j\omega_1 k}$.

Ans-4 $a_0 = 0$; $a_{-1} = -j$, $a_{-2} = -2j$, $a_{-3} = -3j$

Ans-5 (i) $x[n] = -j\frac{1}{2} e^{j3\Omega_0 n} + \frac{1}{2} e^{j4\Omega_0 n} + \frac{1}{2} e^{j20\Omega_0 n} + j\frac{1}{2} e^{j21\Omega_0 n}$; $\Omega_0 = \pi/12$.

(ii) $x[n] = \frac{1}{2} + \frac{1}{4} e^{j\Omega_0 n} + \frac{1}{4} e^{j7\Omega_0 n}$; $\Omega_0 = \pi/4$

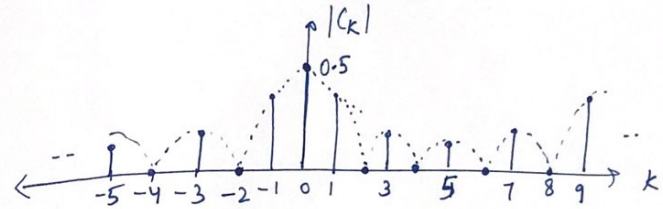
Ans-6 (i) $c_k = \frac{1}{4}$ $\forall k$.

(ii) $c_0 = 3/2$; $c_1 = \frac{-1+j}{2}$

$c_2 = -1/2$; $c_3 = \frac{-1-j}{2}$

Ans-7 $c_k = \frac{1}{10} e^{-jk2\pi/5} \cdot \frac{\sin(k\pi/2)}{\sin(k\pi/10)}$;

$k = 0$ to 9.



Ans-8 $x[n] = 2 + 4\sin[(4\pi n/5) + 2\pi/3] + 2\sin[8\pi n/5 + \frac{5\pi}{6}]$

Ans-9 $x_1[n] \cdot x_2[n] \xleftrightarrow{F.S} 6$ $\forall k$.

Ans-10 $H(e^{j0}) = H(e^{j\pi}) = 0$;

$H(e^{j\pi/2}) = 2e^{j\pi/4}$ & $H(e^{3j\pi/2}) = 2e^{-j\pi/4}$

Ans-11 (i) FS coefficient of $g[n] = \frac{1}{10}[1 - e^{-j(2\pi/10)8k}]$

(ii) $a_k = \frac{1/10[1 - e^{-j(2\pi/10)8k}]}{1 - e^{-j(2\pi/10)k}}$

Ans-12 $y(t) = 0$.